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PREFACE

Vedic Mathematics is a collection of ancient tricks and techniques to execute arithmetic operations quickly and more efficiently. Vedic Math comes from the Vedas, more specifically the Atharva Veda. It was revived by Indian mathematician **Jagadguru Shri Bharati Krishna Tirthaji** between 1911 and 1918. He then published this work in a book called Vedic Mathematics in 1965. It comprises 16 sutras (formulae) and 13 sub sutras. Vedic Math is an ancient technique that simplifies multiplication, divisibility, complex numbers, squaring, cubing, square roots, cube roots, recurring decimals, and auxiliary fractions. Vedic maths provides answers in one line, as opposed to the several steps of traditional mathematics. There are six Vedanganas. The Jyotish Shastra is one of the six. Vedic Math forms part of this Jyotish Shastra. Vedic maths consists of 3 segments or 'skandas' (branches). The beauty of Vedic Math lies in its simplicity; all calculations can be done on pen and paper. The approach to solve problems stimulates and sharpens the mind, memory, and focus. It improves creativity and promotes innovation.

PRELIMINARIES

In this section we collect some basic Terms and definitions.

Nikhilam Sutra: it can be used as a shortcut to multiply numbers, divide numbers in faster approach. In English it is translated as All from 9 and last from 10.

Urdhva -Tiryak: is a generalized sutra which can be applied for multiplication of two numbers having p digits. The meaning of Urdhva Tiryak sutra is simple: “Urdhva means vertically Tiryak means cross wise”.

Factorization: Factorization is defined as a form of “Reversed multiplication” and as a particular application of division.

Auxiliary: it is a helper which convert vulgar fractions into Recurring or infinite decimals.

Fractions: it represents any number of equal parts it is a part of whole it consists of a numerator which is present above a line or before a Slash like ½ and non-zero denominator, which is displayed before or after that line.

Auxiliary Fractions: The technique we have used for calculating recurring decimals in the previous lessons can be modified and used to do some kinds of regular division also. The method uses a computational device called auxiliary fractions to perform the division.

CHAPTER 1

MULTIPLICATION

Multiplication of numbers plays a very important role in most computations. Vedic maths offers two main approaches to multiplying 2 numbers. These are the

- NIKHILAM SUTRA
- URDHAVA-TIRYAK SUTRA

In the NIKHILAM method we subtract a common number, called the base, from both the numbers. The Multiplication is then between the difference so obtained. URDHAVA TIRYAK is General method of multiplication in Vedic Maths. URDHAVA TIRYAK method which uses cross multiplication.

1.1 MULTIPLICATION By NIKHILAM SUTRA

In Vedic mathematics there are number of multiplication methods based on different Vedic Sutras. To begin with, we discuss below the methods based on 'NIKHLAM Sutra' which reads: "NIKHILAM NAVATSCHCHRAMAM DASHTAH"(All from 9, last from 10). This method is known as “NIKHILAM Sutra method”.

This method helps us to multiply numbers which are nearer to the powers of 10. It helps to multiply big numbers with less number of computations. When any number is multiplied by one there is no change. For example,

4x1=4,

271 × 1 = 271

When one number is multiplied by another then it is increased and becomes further away from one. For example, when 2 is multiplied by 3 it becomes 6 which is further away from 1 than 2.

1.1.1 ILLUSTRATION

Let us first take up a very easy and simple illustrative example, the multiplication of single-digit numbers above 5 and see how this Can be done without previous knowledge of the higher multiplications of the multiplication table. Suppose we have to multiply 9 by 7.

I. Take a base for our calculations. That power of 10 which is nearest to the numbers to be multiplied. In this case 10 itself is that power.

II. Put the numbers 9 and 7 above and below on the left-hand side.

III. Subtract each of them form the base (10) and write down the remainders (1 and 3) on the right-hand side with a connecting minus sign (-) between them, to show that the numbers to be multiplied are both of them less than 10.

IV. The product will have two parts, one on the left side and one on the right. A vertical dividing line may be drawn for the purpose of demarcation of the two parts.

Left hand side digit can be arrived at in one of 4 ways.

a) Subtract the base 10 from the sum of the given numbers. And put (l6-10) that is 6, as the left-hand Part of the answer.

b) Subtract the sum of the two deficiencies from the base.

c) Cross-subtract deficiency 3 on the second Row from the original number 9 in the First row. And you find that you have got 6 again.

d) Cross subtracts in the converse way .and you get 6 again as the left-hand side portion of the required answer.

Now vertically multiplies the two figures. The product is 3. And this is the right-hand side portion of the answer. Thus 9×7=63

(10)

9-1

<u>7-3</u>

6/3

1.1.2 MULTIPLES AND SUB-MULTIPLES

Yes, but in all these cases, we find both the multiplicand and the multiplier, or at least one of them, very near the base taken in each case; and this gives us a small multiplier and thus renders the multiplication Very easy. The needed solution for this purpose is furnished by a small Upasütra' (or sub-formula) which is so-called because of its practically axiomatic character.

when neither the multiplicand nor the multiplier is sufficiently near a convenient power of 10 which can suitably serve as a base, we can take a convenient multiple or sub-multiple of a suitable base, as our "working base, perform the necessary operation With its aid and then multiply or divide the result proportionately, that is in the same proportion as the original base may bear to the working base actually used by us. A concrete illustration will make the modus operandi clear.

Suppose we have to multiply 41 by 41. Both these numbers are so far away from the base 100 that by our adopting that as our actual base, we shall get 59 and 59 as the deficiency from the base. And thus the consequent vertical multiplication of 59 by 59 would prove too cumbrous a process to be permissible under the Vedic system and will be positively inadmissible.

We therefore, accept 100 merely as a theoretical base and take submultiple 50 which is conveniently near 41 and 41 as our working basis, Work the sum up accordingly and then do the proportionate multiplication or division, for getting the correct answer. Thus:

i. We take 50 as our working base.

ii. By cross-subtraction, we get 32 on left-hand side.

iii. As 50 is a half of 100, we therefore divide 32 by 2 and put 16 down as the real left-hand side portion of the answer.

iv. The right-hand side portion 81 remains unaffected.

v. The answer therefore is 1681.

100/2=50

41-9

41-9

2) 32/81

16/81

Examples for multiplication of numbers below the base (10)

Sr.No	Exercise	Answer
1	97 × 98	8051
2	96 × 98	9408
3	97 × 92	8924
4	88 × 98	8624
5	87 × 840	73080
6	97 × 93	9021
7	88 × 89	7832
8	85 × 8.9	756.50
9	97 × 320	31040
10	9.8 × 0.95	9.31
11	8.4 × 0.089	0.7476
12	8.8 × 9.7	85.36
13	7.6 × 8.7	66.12
14	0.0092 × 0.0092	0.00008464
15	8.7 × 93	809.10
16	0.77 × 98	75.46
17	86 × 0.97	83.42
18	9.6 × 79	758.40
19	0.085 × 93	7.905
20	6.9 × 8.8	60.72

1.1.4 corollary

The corollary deals with the squaring of numbers. A few elementary Examples will suffice to make its meaning and application clear: Suppose we have to find the square of 9. The following will be the Successive stages in our mental working:

i. We should take up the nearest power of 10, that is 10 itself as our Base.

ii. As 9 is 1 less than 10, we should decrease it still further by 1 and set 8 down as our left-side portion of the answer.

iii. And, on the right hand, we put down the square of that deficiency 1^2.

iv. Thus $9^2= 81$.

Now, let us take up the case of 8^2. As 8 is 2 less than 10, we lessen it still further by 2 and get 8-2, that is 6 for the left-hand and putting $2^2=2=4$ on the right-hand side, we say $8^2=64$.

In exactly the same manner, we say

$7^2= (7-3)/3^2=4/9$

$6^2= (6-4)$ and $4^2=2/_16=3/6$

$5^{2=}$ (5-5) and 5^2-$0/_2$=25; and so on.

1.1.5 corollary

The corollary is applicable only to a special case under the first corollary, i.e., the squaring of numbers ending in 5 and other cognate numbers. It relates to the squaring of numbers ending in 5, example say ,15. Here, the last digit is 5; and the “previous” one is 1. So, one more than that is 2. Now, the Sutra in this context tells us to multiply the previous digit by one more than itself, that is by 2. So, the left-hand side digit is 1x2; and the right-hand side is the vertical multiplication-product, that is 25 as usual.

Thus 15^2=1x2/25=2/25

Similarly,

25^2=2×3/25=6/25;

35^2=3×4/25=12/25;

55^2=5×6/25=30/25;

65^2=6×7/25=42/25;

75^2=7×8/25=56/25;

1.2 MULTIPLICATION BY URDHAVA TIRYAK SUTRA

In the previous section we discussed about NIKHILAM sutra which is applicable to the special cases of multiplication. Now we deal with another sutra which is applicable to all cases of multiplication. Urdhva tiryak sutra is the general formula applicable to all cases of multiplication and also in the division of a large number by another large number. It means "Vertically and cross wise.". it consists of vertical and cross multiplication between the various digits and gives the final result in a single line.

1.2.1 ILLUSTRATION

Suppose we have to multiply 12 by 13.

1. We multiply the left-hand-most digit 1 Of the multiplicand vertically by the left- hand most digit I of the multiplier, get their product 1 and set it down as the Left-hand-most part of the answer;
2. We then multiply 1 and 3, and 1 and 2 cross-wise, add the two, get 5 as the sum and set it down as the middle part of the answer;
3. We multiply 2 and 3 vertically, get 6 as their product and put it down as the last the right-hand-most part of the answer.

   Thus 12x13=156.

1.2.2 <u>Algebraic Principle</u>

If two numbers are expressed in the form (ax+b) and (cx+d), then the multiplication result is ($acx^2 + x(ad+bc) + bd$). Notice that coefficient of x^2 is ac which is the result of vertical multiplication of a and c. The coefficient of x is the result of cross-wise multiplication of a, d and b, c and addition of the two products. The independent term bd is the result of vertical multiplication of absolute terms b, d.

If we substitute x with 10, we notice that the above expression is applicable to Arithmetical numbers as explained in the Urdhva-Tiryagbhyam Sutra above. We can expand the same principle to numbers with more than 2 digits: With x = 10, we can express two 3-digit numbers as (ax2 + bx + c) and (dx2 + ex + f). Multiplication of such two numbers gives us:

(ax2 + bx + c) * (dx2 + ex + f) = adx4 + x3(ae+bd) + x2(af+be+cd) + x(bf+ce) + cf.

1.2.3 <u>Facts</u>:

1. That the coefficient of X^4 is got by the vertical multiplication of the first digit from the left side;
2. that the coefficient X^3 is got by the cross-wise multiplication of the first two digits and by the addition of the two products;
3. that the coefficient of X^2 is obtained by the multiplication of the first digit of the multiplicand by the last digit of the multiplier, of the middle one by the middle one and of the last one by the first one and by the addition of all the three products;
4. that the coefficient of x is obtained by the cross-wise multiplication of the second digit by the third one and conversely by the addition of the two products;
5. that the independent term results from the vertical multiplication of the last digit by the last digit.

1.2.4 EXAMPLES:

1. 1 1 1

   1 1 1

   12321

2. 108

   108

   10 60 4

   1 6

   11 66 4

3. 5 8 2

   2 3 1

   10 1 3 42

4. 8 7 2 6 5

   2 3 1 1 7

   2 4 7 8 7 2 7 5 7 5

   3 2 3 9 6 2 4 3

   2 8 0 2 6 9 0 0 0 5

1.2.5 Algebraic example:

A. a + b

a+9b

$a^2+10ab+b^2$

Note: If and when a power of x is absent, it should be given a zero coefficient; and the work should be carried on exactly as before.

1.2.6 MISCELLANEOUS EXAMPLES

There being so many methods of multiplication one of them the Urdhva-Tiryak being perfectly general and therefore applicable to all cases and the others the Nikhilam, the Yavadunam etc. being of use in certain special cases only, it is for the student to think of and weigh all the possible alternative processes available, make up his mind as to the simplest method in each particular case and apply the formula prescribed therefore now conclude this with a number of miscellaneous examples by the various alternative methods;

1) 73x37

- By Urdhva-Tiryak rule,

$$\begin{array}{r} 73 \\ \underline{37} \\ 2181 \\ \underline{52} \\ \underline{=2701} \end{array}$$

- By the same method but with the use of the vinculum,but the former is better.

2) 94×81

- By Urdhva-Tiryak,

$$\begin{array}{l} 94 \\ \underline{81\quad} \\ 7614 \end{array}$$

- By ibid (with the use of the vinculum). Evidently the former is better but The NIKHILAM method is still better.

CHAPTER 2

FACTORIZATION

Factorization is defined as a form of "Reversed multiplication" and as a particular application of division. The Factorization of a quadratic expression into its component binomial factors can be done by the following methods Such as;

- Factorization of simple Quadratics
- Factorization of "Harder" Quadratics

2.1 Factorization Of Simple Quadratics

Factorization using Vedic Mathematics is done by the combination of 2 sutras.

1. Anurupyena (Proportionality).
2. Adyamadyenantyamantya (1^{st} by 1^{st} and last by last)

Both the above-mentioned sutras can be worked out as follows:

Let the quadratic expression be $2x^2+5x+2$;

I. Split the middle coefficient into two such parts that the ratio of the first coefficient to that first part is the same as the ratio of that second part to the last coefficient. Thus, in the quadratic $2x^2+5x+2$, the middle term 5 is split into two such parts 4 part and 1 that the ratio of the first coefficient, to the first part of the middle coefficient, i.e. 2:4 and the ratio of the second part of the last coefficient ,i.e. 1:2 are the same. Now, this ratio,i.e. $x+2$ is one factor.

II. And the second factor is obtained by dividing the first coefficient of the quadratic by the first coefficient of the factor already found and the last coefficient of the quadratic by the last coefficient of that factor. In other words the second Binomial factor is obtained thus: $2X^2/2 + 2/2 = 2X + 1$ .Thus we say :$2x^2+5x+2= (x+2)(2x+1)$.

2.1.1 Note

The middle coefficient [which we split up above into (4+1)] may also be split up into 1 +4, that the ratio in that case is 2x+1 and that the other Binomial factor according to the above-explained method is x +2. Thus, the change of sequence in the splitting up of the middle term makes no difference to the Adyamadyenantyamantya.

2.1.2 Examples

I. $7x^2-6x-1= (x-1)(7x+1)$

II. $6x^2+37x+6= (x+6)(6x+1)$

III. $4x^2+12x+5= (2x+1)(2x+5)$

IV. $3x^2+13x-30 = (x+6)(3x-5)$

V. $2x^2+5x-3= (x+3)(2x-1)$

2.2 Factorisation of "Harder" Quadratics

The class of quadratic expression known as the Homogeneous Expressions of second degree, wherein several letters x,y,z etc. Which can be tackled by means of "Adyamadyena Sutra" and the sub-sutra The "LopanaSthapana" which consists of only one compound word, this removes the whole difficulty and makes the factorization of quadratic of this type as easy as the simple as that of the ordinary quadratic.

2.2.1 Illustration

Suppose we have to factorize the following long quadratic:

$2x^2+6y^2+3z^2+7xy+11yz+7zx$

1. We first eliminate z by putting z=0 and retain only x and y and factorize the resulting ordinary quadratic in x and y with the Adyam Sutra;

2. We then similarly eliminate y and retain only x and z and factorize the simple quadratic in x and z;
3. With these two sets of factors before us, we fill in the gaps caused by our own deliberate elimination of z and y respectively.

And that gives us the real factors of the given long expression. The procedure is an argumentative one and is as follows:

If z=0, then E (the given expression) = $2x^2+7xy+6y^2$

$$= (x+2y)(2x+3y)$$

Similarly, if y=0, then

$$E=2x^2+7xz+3z^2 = (x+3z)(2x+z)$$

Therefore, Filling in the gaps which we ourselves had created by leaving out z and y we say :

$$E=(x+2y+3z)(2x+3y+z).$$

2.2.2 Examples

I. $2x^2+2y^2+5xy+2x-5y-12= (x+3)(2x-4)$ and also $(2y+3)(y-4)$

Therefore, $E=(x+2y+3)(2x+y-4)$

II. $3x^2+8xy+4y^2+4y-3=(x-1)(3x+3)$ and also

Therefore, $E=(x+2y-1)(3x+2y+3)$

2.2.3 Note

We could have eliminated x also and retained only y and z and factorized the resultant simple quadratic. That would not, however, have given us any additional material but would have only confirmed and verified the answer we had already obtained. Thus, when 3 letters x, y and z are there, only two eliminations will generally suffice.

CHAPTER 3

AUXILIARY FRACTION

There are certain Vedic process, however, by which with the aid of what we call auxiliary fractions, the burden of the subsequent operations is also considerably lightened and the work is splendidly facilitated. The method uses a computational device called auxiliary fractions to perform the division.

3.1 Divisors Ending with ‘9’

Let us start with the case where the divisor ends with 9. E.g., 19, 29, 39, 89 etc. General Steps

Add 1 to the denominator and use that as the divisor. Remove the zero from the divisor and place a decimal point in the numerator at the Appropriate position Carry out a step-by-step division by using the new dividend

Every division should return a quotient and a remainder and should not be a decimal division Every quotient digit will be used without any change to compute the next number to be Taken as the dividend. In all other cases, the quotient digit would be altered by a pre-defined Method which will be explained in each case.

3.1.1 Example 1:

Let us now see the details for 5 / 29 Steps

- Add 1 to the divisor to get 30
- The modified division is now 5 / 30
- Remove the zero from the denominator by placing a decimal point in the numerator Giving the modified division as 0.5 / 3
- We will not write the final answer as 0.16666 but
- We will carry out a step-by-step division explained below

Steps for step-by-step division

- Divide 5 by 3, to get a quotient (q) = 1 and remainder (r) = 2
- Write down the quotient in the answer line and the remainder just below it to its

  Left as shown

  0.5/3= 0 . 1

  2
- Now, the next number for division is 21 which on division by 3 gives q = 7 and r =0 The result now looks as

  0.5/3= 0 . 1 . 7

  2 0
- Repeat the division at each digit to get the final answer to any desired level of Accuracy!! The next few digits are shown below.
- On dividing 7 by 3, we get q = 2 and r = 1. The result now looks as

  0.5/3= 0. 1 7 2

  2 0 1
- On dividing 12 by 3, we get q = 4 and r = 0. The result now looks as

  0.5/3= 0 . 1 7 2 4

  2 0 1 0
- The final result up to 8 digits of accuracy is

  0.5/3= 0 . 1 7 2 4 1 3 7 9

  2 0 1 0 1 2 2 0

And the result of 5/29 = 0.17241379.

3.1.2 Notes:

For all numbers ending with 3, we can convert them to numbers ending with 9 By multiplying by 3. E.g., if the divisor is 13, we can convert it to 39 by multiplying by 3. Hence, 3/13 = 9/39, 5/23 = 15/69 Similarly, for numbers ending with 7, we can multiply by 7 to get a number ending with 9.

3.2 Divisors Ending with '1'

Now, let us consider the fractions where the divisors are numbers ending with 1.

E.g., 21, 31, 51, 71 etc…

General steps

- Subtract 1 from the numerator
- Subtract 1 from the denominator and use that as the divisor
- Remove the zero from the divisor and place a decimal point in the numerator at the appropriate position
- Carry out a step-by-step division by using the new divisor
- Every division should return a quotient and remainder and should not be a decimal division
- Subtract the quotient from 9 at each step to form the next dividend

3.2.1 Example 1:

Let us now see the details for 4 / 21

Steps

- Subtract 1 from the numerator
- Subtract 1 from the divisor to get 2
- The modified division now is 3 / 20
- Remove the zero from the denominator by placing a decimal point in the numerator
- The modified division is 0.3 / 2
- We will not write the final answer as 0.15 but, we will carry out a step-by-step division as explained below.

Steps for step-by-step division

- Divide 3 by 2, to get a quotient (q) = 1 and remainder (r) = 1
- Write down the quotient in the answer line and the remainder just below it to it's Left as shown,

0.3/2=0 . 1
1

- Subtract the quotient (1) from 9 and write it down

. 8
.
0.3/2=0 . 1
1

- Now, the next number for division is 18 and not 11.
- On dividing 18 by 2, we get the q = 9 and r = 0. The result now looks as

. 8
.
0.3/2=0 . 1 9
1 0

- Subtract the quotient (9) from 9 and write it down

. 8 . 0
. .
0.3/2=0 . 1 9
1 0

- Now, the next number for division is 0 and not 9.
- Repeat the division as explained before.
- On dividing 0 by 2, we get q = 0 and r = 0. The result now looks as

. 8 . 0 . 9
. . .
0.3/2=0 . 1 9 0
1 0 0

- The final result up to 8 digits of accuracy is

. 8 . 0 . 9 . 5 .2 . 3 . 8 . 0

. . . .

0.3/2= 0 . 1 9 0 4 7 6 1 9

1 0 0 1 1 0 1 0

- and the result of 4/21 = 0.19047619

3.2.2 Notes:

For all numbers ending with 3, we can convert them to numbers ending with 1 by multiplying by 7.

E.g., If the divisor is 13, we can convert it to 91 by multiplying by 7.

Hence, 3/13 = 21/91, 5/23 =35/161

Similarly, for numbers ending with 7, we can multiply by 3 to get a number ending with1. E.g., if the divisor is 17, we can convert it to 51 by multiplying by 3.

Hence,

5/7 =15/21

4/17 =12/51

5/27 =15/81

Once the methods are clear for divisors ending with 9 and with 1, we can move over to other divisors ending with 8, 7 and 6.

3.3 Other Divisors

In the sections given above, we have seen divisors ending with 1, 6, 7, 8 and 9. What do we do with divisors ending with 2, 3,4and 5? We can convert such divisors, by a suitable multiplication, to divisors for which we have seen Vedic techniques. Use the table given below for guidance.

Sr.No	Divisor ending with	Multiply by	Ending with	After Multiplication
1.	2	5 4	13/42 17/22	65/210 68/88
2.	3	3 7	13/43 13/43	39/129 91/301
3.	4	5 7	13/34 13/34	65/170 91/238
4.	5	2	13/235	26/470
5.	6	3 5	5/26 5/26	15/78 25/130

We can see from this table that if a proper digit is used to multiply the numerator and the denominator, we can obtain a fraction wherein we can apply one of the known techniques and derivate answer quickly. Sometimes, we may have to carry out the multiplication twice to get it in a standard form.

CONCLUSION

Vedic mathematics is the source of the modern mathematics, that we are studying now in schools and universities. We can discover numerous helpful methods to solve the mathematics problem through Vedic Mathematics. Vedic Maths is an excellent practice that helps to sharpen our brain and improves the calculation power. These techniques are derived from 16 sutras (verses) in the Vedas, which are Thousands of years old and among the earliest literature of ancient Hindus in India. They are an endless source of knowledge and wisdom, providing practical knowledge in all spheres of life. Here we have discussed some of these techniques such as Multiplication, Division and Auxiliary Fractions using Vedic mathematics.

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